To what extent does defense R&D contribute to technological spillovers and economic growth, and how does this differ between countries that produce defense technology and those that rely on imports?

Victor Cho

Lewis & Clark College
Portland, Oregon
Department of Economics
Submitted in partial fulfillment for the requirements of the Degree of Bachelor of Arts
April 29, 2026

Abstract: Rising geopolitical tensions have led to a substantial increase in global defense spending, often viewed as a diversion of resources from productive economic use. This paper examines the conditions under which defense spending contributes to technological spillovers and long-run economic growth, with a particular focus on the role of domestic research and development (R&D). The analysis distinguishes between countries that produce defense technologies and those that rely on imports, emphasizing the importance of domestic production capacity.

Using data from the SIPRI Arms Transfers Database, countries are classified as net exporters or importers of defense technologies to capture differences in defense-industrial capability. An empirical framework is employed to evaluate whether defense R&D has a stronger effect on economic outcomes in countries with established domestic production systems.

The results indicate that defense spending supports economic growth primarily when it is linked to domestic innovation and industrial development. Countries that rely heavily on imports capture fewer technological spillovers, suggesting that the economic benefits of defense investment depend less on its magnitude and more on its composition. These findings highlight the importance of domestic capability in translating defense spending into sustained economic gains.

Keywords: Defense R&D; Technological Spillovers; Economic Growth; Military Spending; Defense-Industrial Base; Arms Trade; Innovation; Domestic Production; Productivity; SIPRI

JEL Codes: H56, O38, O33, O47, F52

1. Introduction

The current global environment is not particularly optimistic. Conflicts in the Middle East, rising tensions in East Asia, and increasing geopolitical competition across regions have pushed countries to significantly expand defense spending. At first glance, this trend appears entirely negative, war, instability, and the diversion of resources away from productive economic use. And in many cases, that interpretation is correct. However, stopping the analysis there misses an important part of the story. If anything, history suggests that periods of conflict often coincide with bursts of technological progress. In other words, even in a darker environment, there are still glimpses of economic opportunity.

Economists have long debated whether military spending promotes or hinders economic growth. A large portion of the literature finds that defense spending can crowd out productive investment, particularly in developing economies where resources are limited and spending is often directed toward imports. However, more recent research emphasizes that the composition of defense spending matters. Defense research and development (R&D), unlike general military expenditure, is closely linked to technological innovation and can generate spillover effects into the civilian economy. Many foundational technologies of the modern economy including semiconductors, satellite systems, and the internet originated from military-funded research before diffusing into private industry.

This paper builds on that distinction and takes a more conditional view. The central question of this paper is therefore straightforward: does defense R&D actually make countries economically stronger, and under what conditions does that happen? The answer, as this paper argues, is not universal. Defense spending is not inherently good or bad for growth, it depends on whether it builds domestic capability or simply finances foreign production. That distinction turns out to matter a lot.

Specifically, this paper argues that defense spending contributes to long-run economic growth only when it is tied to domestic innovation and industrial capacity. Simply spending more on the military does not guarantee economic benefits. In fact, in many cases, it does the opposite. But when defense spending is directed toward domestic R&D and production, it can act as a catalyst for technological development, strengthening both military capability and civilian industries. This leads to a key difference across countries. Nations such as the United States, Israel, and South Korea have developed strong domestic defense industries and are able to capture the technological spillovers generated by defense investment. In contrast, countries that rely heavily on importing weapons systems despite sometimes spending large amounts often fail to internalize these benefits. In simple terms, buying technology is not the same as building it.

To examine this relationship, this paper combines macroeconomic theory with empirical evidence. Using data from the SIPRI Arms Transfers Database, countries are classified based on their position as net exporters or importers of defense technology, which serves as a proxy for domestic production capability. The analysis then evaluates whether defense R&D has a stronger impact on economic outcomes in countries with established defense-industrial systems. In addition, firm-level evidence is used to illustrate how defense spending affects expected profits and market behavior, highlighting the financial channel through which government spending translates into economic activity.

The remainder of the paper proceeds as follows. Section 2 reviews the literature on defense spending, innovation, and economic growth. Section 3 presents the theoretical framework linking defense R&D to productivity and spillover effects. Section 4 provides the empirical analysis using cross-country data and financial market evidence. Section 5 concludes.

2. Literature Review

2.1 Theoretical literature

The relationship between defense spending and economic growth has been studied through multiple theoretical frameworks, with no clear consensus on its overall impact. In the neoclassical tradition, military expenditure is typically modeled as a public good that provides national security but comes at an opportunity cost. Resources allocated to defense reduce the amount available for private consumption and investment, potentially lowering long-run economic growth. In this framework, the government chooses an optimal level of defense spending by balancing security benefits against economic costs (Dunne and Uye 2010).

However, this perspective has been expanded in more recent theoretical work, particularly within endogenous growth models. Unlike traditional models where technological progress is treated as exogenous, endogenous growth theory emphasizes that innovation and research investment are central drivers of long-run economic growth (Romer 1990; Aghion and Howitt 1992). Within this framework, defense spending, specifically defense research and development (R&D) can influence economic performance by affecting the rate of technological progress.

Macroeconomic modeling also provides further insight into the interaction between military and civilian sectors. Torres (2020) develops a two-sector dynamic stochastic general equilibrium (DSGE) model in which military and non-military production compete for resources such as capital and labor. The model shows that defense spending does not directly increase output, but instead affects economic activity through general equilibrium mechanisms, including resource allocation and productivity effects. Importantly, technological improvements in the defense sector can generate positive spillovers into the civilian economy by increasing overall efficiency and freeing resources for non-military production.

At the same time, theoretical models also highlight potential trade-offs. Because total resources devoted to research are limited, increased defense R&D may crowd out civilian innovation if resources are reallocated toward military projects. As a result, the net effect of defense investment depends on the balance between positive spillovers and negative crowding-out effects. This ambiguity is central to the ongoing debate and motivates the need for empirical analysis.

2.2 **Empirical literature**

The empirical literature on defense spending and economic growth is extensive but highly mixed. A large number of cross-country studies find that military expenditure has either a negative or statistically insignificant effect on economic growth, particularly in developing economies. Dunne and Uye (2010), in a survey of over one hundred studies, report that only a minority find a positive relationship, while many show negative or neutral effects. Similarly, Knight, Loayza, and Villanueva (1996) find that high levels of military spending reduce growth by diverting resources away from productive investment.

These results are often explained by the composition of military spending. In many developing countries, defense budgets are heavily concentrated in personnel costs or imports of foreign weapons systems, which generate limited domestic economic benefits. As a result, defense spending in these contexts does not translate into technological development or productivity gains.

In contrast, a growing body of empirical research focuses specifically on defense R&D and its role in innovation. Block and Keller (2011) argue that U.S. technological development has historically been driven by mission-oriented government programs, particularly through defense institutions such as DARPA, which played a central role in shaping industries like semiconductors, aerospace, and computing. Similarly, Mazzucato (2013) emphasizes the role of

the state as an “entrepreneurial actor,” highlighting that key technologies such as GPS and the internet originated from publicly funded defense research.

Firm-level evidence further supports the existence of positive spillovers. Draca (2013) finds that increases in defense-related procurement are associated with higher levels of patenting and innovation in private-sector firms. Moretti, Steinwender, and Van Reenen (2019) provide additional evidence of crowding-in effects, showing that government-funded R&D stimulates private-sector investment and technological development. These findings suggest that defense R&D can contribute to productivity growth by strengthening innovation systems.

However, this optimistic view is not universally accepted. McDonough (2012) argues that defense R&D may also generate negative effects by diverting resources away from civilian sectors and increasing the cost of skilled labor. Harutyunyan (2017) similarly emphasizes that military R&D operates through both positive and negative channels, including spillovers, crowding-out, and institutional inefficiencies. Guellec and van Pottelsberghe (2004) also find that military R&D may have weaker or even negative effects on productivity compared to civilian R&D. These findings suggest that the economic impact of defense R&D depends critically on institutional context and the structure of the economy.

Table 1: Summary of Empirical Literature on Defense Spending and Growth

Study	Sample	Method	Key Findings

Dunne & Uye (2010)	100+ studies (cross-country)	Literature review	Majority find negative or insignificant effects of military spending on growth
Knight, Loayza & Villanueva (1996)	Cross-country	OLS regression	High military spending reduces growth via lower capital accumulation
Leonard (1971)	U.S. firms (1960s)	Regression	Government R&D weakened industrial productivity (crowding out)
Guellec & van Pottelsberghe (2004)	OECD countries	Panel regression	Military R&D negative, civilian R&D positive for productivity
Draca (2013)	U.S. firms	Firm-level regression	Defense procurement increases patents and private R&D
Moretti, Steinwender & Van Reenen (2019)	OECD industries	Quasi-experimental	$1 defense R&D generates $2–$5 private R&D (crowding-in)
Harutyunyan (2016)	OECD countries	Panel model	Mixed effects depending on civilian R&D capacity
Yakovlev (2007)	Cross-country	Regression	Military spending negative unless country exports arms

Fredericksen & Looney (1982)	Developing vs developed	Regression	Positive effects only in industrialized/resource-rich economies

2.3 Synthesis and implications

Taken together, the literature suggests that the economic effects of defense spending depend heavily on how resources are allocated between military and civilian uses. Studies focusing on aggregate military expenditure tend to find weak or negative effects on economic growth, particularly in countries that rely on imported defense systems. In contrast, research focusing on defense R&D highlights the potential for technological spillovers and productivity gains, especially in economies with strong innovation systems.

This distinction points to a key mechanism: the ability of a country to internalize the benefits of defense investment depends on its domestic production and innovation capacity. Countries that develop their own defense industries are more likely to capture spillover effects, while countries that rely on imports may fail to do so. Building on this insight, the next section develops a theoretical framework linking defense R&D to technological progress and economic growth, which is then tested empirically using cross-country data.

3. Theoretical Framework

This section develops a theoretical framework to analyze how defense research and development (R&D) may influence economic growth. The central idea is that the impact of defense R&D depends on its ability to generate technological spillovers and contribute to domestic production capacity.

Economic output is modeled using a standard Cobb–Douglas production function

Output is defined using a standard Cobb–Douglas production function:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}$$

where Y_t represents national output, K_t is the capital stock, and L_t is labor input. The key variable is A_t, which captures total factor productivity and reflects the level of technology in the economy. Unlike in standard neoclassical models, technology is not treated as an exogenous residual but is instead determined by investment in research and development across sectors.

Technological progress is modeled as:

$$A_t = A_0 + \gamma R_{d,t} + \delta R_{c,T}$$

where $R_{d,t}$ denotes defense R&D and $R_{c,t}$ denotes civilian R&D. The parameters γ and δ represent their respective contributions to technological progress. This formulation captures the central theoretical argument of the paper: defense-sector investment can influence economic growth through its effect on innovation and productivity. Substituting this expression into the production function implies that output depends not only on factor accumulation, but also on the composition of R&D investment across sectors.

A key feature of the model is that total research resources are limited. This constraint can be expressed as:

$$R_{d,t} + R_{c,t} = R$$

This introduces a trade-off between defense and civilian R&D. An increase in defense R&D may generate technological gains in defense-related sectors, but it may also divert resources away from civilian innovation. As a result, the overall effect of defense R&D on productivity is theoretically ambiguous.

Within this framework, defense R&D can affect economic output through two primary channels. The first is a productivity channel, in which defense-related research contributes to technological progress and raises total factor productivity:

$$R_{d,t} \rightarrow A_t \rightarrow Y_t$$

In this channel, knowledge generated through defense R&D can spill over into other sectors of the economy, improving efficiency and increasing output.

The second channel operates through capital accumulation. Defense R&D may increase expected profitability and investment, leading to an expansion of the capital stock. Firm profits can be expressed as:

$$\pi_t = P_t Y_t - w_t L_t - r_t K_t$$

where π_t denotes profits, P_t is the price of output, Y_t is total output, W_t is the wage rate, L_t is labor input, and r_t is the rental rate of capital. An increase in defense R&D raises productivity and output, which increases revenues and, holding input costs constant, leads to higher profits.

Higher expected profits increase firm valuation and incentivize investment, resulting in an expansion of the capital stock. This provides a second transmission mechanism through which defense R&D affects economic output:

$$R_{d,t} \rightarrow V_t \rightarrow I_t \rightarrow K_t \rightarrow Y_t$$

An important implication of this framework is that the effects of defense R&D are not uniform across countries. Economies with domestic defense-industrial capability are more likely to internalize technological gains and incorporate them into local production processes. In contrast, countries that rely heavily on imported defense technologies may not experience the same benefits, as defense spending does not translate into domestic innovation or productivity growth.

This leads to two key theoretical predictions. First, defense R&D is expected to have a positive effect on economic output through its contribution to technological spillovers and productivity growth. Second, this effect is expected to be stronger in countries with domestic production capability. At the same time, the presence of resource constraints and potential crowding-out effects implies that the magnitude of this relationship may be limited, and the overall impact of defense R&D may appear weak or ambiguous when averaged across all countries.

4. Data and Empirical Strategy

4.1 Data Sources and Variables

The empirical analysis in this paper uses an unbalanced panel dataset covering 35 OECD and OECD-comparator countries over the period 2000 to 2022. The unit of observation is the country-year, resulting in approximately 660 observations depending on data availability. The dataset combines information from multiple international sources in order to capture economic performance, innovation activity, and structural differences in defense production.

The research and development variables are obtained from the OECD Main Science and Technology Indicators database. These include total R&D expenditure as a percentage of GDP and civilian R&D expenditure. Defense R&D is not directly reported as a share of GDP, but rather reflects the portion of government R&D allocated toward defense-related activities. As such, defense R&D captures the intensity of innovation directed toward military and strategic technological development rather than overall innovation spending.

Macroeconomic variables are sourced from the World Bank's World Development Indicators. GDP per capita, measured in constant 2017 international PPP dollars, is used as the dependent variable and serves as a standard measure of economic development across countries. Gross fixed capital formation is included as a proxy for capital input, while total labor force is used to capture labor input. Together, these variables reflect the standard components of economic production.

To distinguish between countries with different levels of defense-industrial capacity, a binary exporter variable is constructed using the SIPRI Arms Transfers Database. Countries are classified based on their cumulative net arms trade position over the sample period. Countries with positive net exports are categorized as producers, while countries with net imports are

categorized as importers. This classification allows the analysis to capture whether domestic production capability influences the economic effects of defense-related investment.

All variables are expressed in natural logarithms in the regression analysis. This transformation allows the coefficients to be interpreted as elasticities and helps account for differences in scale across countries. Due to data availability, some variables, particularly defense R&D, have fewer observations, resulting in an unbalanced panel.

To provide a clear overview of the dataset, Table 1 summarizes the definitions, sources, and measurement units of all variables used in the analysis. The table highlights the construction of key variables, including defense R&D, civilian R&D, and the exporter classification derived from arms trade data.

Table 1. Variable Definitions and sources.

Table 1 — Variable Definitions and Sources

Variable	Definition	Source	Units
`ln_Y`	Log of GDP per capita, PPP	World Bank WDI (NY.GDP.PCAP.PP.KD)	Constant 2017 international USD
`ln_rd`	Log of defense R&D spending	Constructed from OECD MSTI	% of GDP
`ln_rc`	Log of civilian R&D spending	OECD MSTI (G_CV)	% of GDP
`ln_gerd`	Log of total gross R&D expenditure	OECD MSTI (G)	% of GDP
`ln_K`	Log of gross fixed capital formation	World Bank WDI (NE.GDI.FTOT.KD)	Constant 2015 USD
`ln_L`	Log of total labor force	World Bank WDI (SL.TLF.TOTL.IN)	Persons
`ln_V`	Log of stock market capitalization	World Bank WDI (CM.MKT.LCAP.GD.ZS)	% of GDP
`exporter`	Dummy equal to 1 if country is a net arms exporter, 0 otherwise	SIPRI Arms Transfers Database, 2000–2022 cumulative net position	Binary
`ln_rd_x_exp`	Interaction of `ln_rd` and `exporter`	Constructed	—
`war`	Dummy for major conflict years: 2001–2003, 2014–2016, 2022	Constructed	Binary

Table 2 presents the descriptive statistics for the variables used in the empirical analysis. The data exhibit substantial variation across countries and over time, which is essential for identifying the relationship between defense R&D and economic performance. GDP per capita, measured in logarithmic form, shows moderate dispersion with a standard deviation of approximately 0.44, reflecting differences in income levels across the sample.

The R&D variables display considerably larger variation. In particular, defense R&D exhibits a wide range of values, with a relatively high standard deviation compared to its mean. This

reflects the fact that defense-related innovation activity differs significantly across countries, especially between those with domestic production capacity and those that rely on imports. Civilian R&D also shows meaningful variation, though it is more concentrated than defense R&D.

The control variables behave as expected. Capital, proxied by gross fixed capital formation, shows substantial dispersion across countries, indicating differences in investment levels and economic structure. Labor, measured as the total labor force, also varies considerably, capturing differences in country size and workforce scale. The financial variable, proxied by stock market capitalization, exhibits notable variability, suggesting differences in financial development and market depth across countries.

Overall, the variation observed in the data supports the empirical strategy by providing sufficient cross-country and time-series differences to estimate the effects of defense R&D. At the same time, the differences in the number of observations across variables indicate the presence of missing data, resulting in an unbalanced panel. This is particularly relevant for defense R&D and financial variables, which are not consistently available across all countries and years.

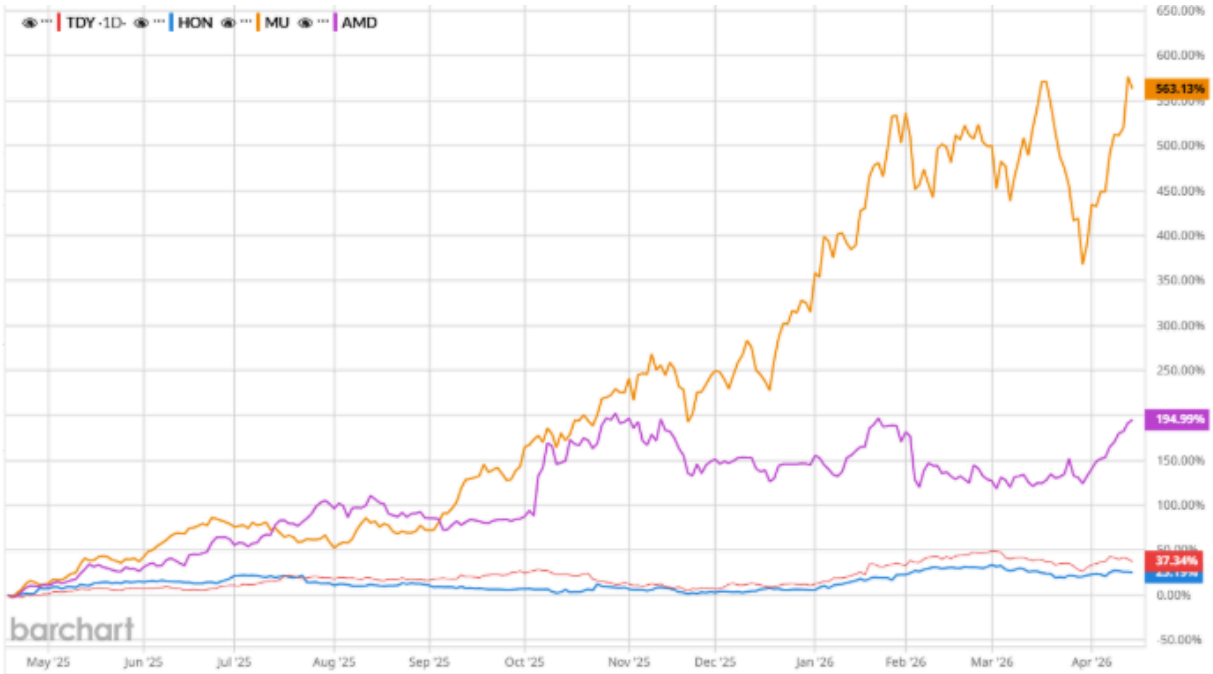
Table 2: Summary statistics

Variable	N	Mean	SD	Min	Max
ln_Y	747.000	10.672	0.442	9.289	11.840
ln_rd	660.000	-3.875	2.259	-13.300	0.491
ln_rc	747.000	0.307	0.685	-1.968	1.539
ln_K	747.000	25.040	1.620	21.511	29.173
ln_L	747.000	15.674	1.514	12.032	18.948
ln_V	549.000	3.763	0.855	-0.023	5.774

Figure 4. Stock Performance of Major Defense Contractors



Figure 5. Stock Performance of Defense-Related and Technology Firms



Figures 4 and 5 provide descriptive evidence of the financial channel outlined in the theoretical framework. Figure 4 shows the stock performance of major defense contractors, while Figure 5 presents firms in related sectors, including technology and industrial production. In both figures, periods associated with increased geopolitical tension or defense activity correspond to rising firm valuations.

This pattern is consistent with the mechanism in which increases in defense R&D raise expected profitability. As expected profits increase, firm valuations rise, reflecting higher anticipated returns. These valuation effects can, in turn, stimulate investment and contribute to capital accumulation, providing an indirect channel through which defense R&D may influence economic output.

At the same time, the variation across firms and sectors highlights that these effects are not uniform. While defense contractors exhibit relatively stable gains, firms in adjacent sectors display greater volatility, suggesting that the transmission of defense-related investment into broader economic activity depends on sectoral linkages and the structure of domestic industries. These descriptive patterns provide motivation for the econometric analysis that follows.

4.2 Econometric Model

The empirical model is designed to examine the relationship between defense research and development (R&D) and economic performance, and to test whether this relationship differs across countries depending on their level of domestic defense-industrial capability. Building on the theoretical framework, the model focuses on both the direct productivity effects of defense R&D and its potential indirect effects through capital accumulation.

The baseline specification follows a standard production-function framework in which output depends on capital, labor, and innovation activity:

$$lnY_t = \beta_0 + \beta_1 lnR_{d,t} + \beta_2 lnR_{c,t} + \beta_3 lnK_t + \beta_4 lnL_t + \alpha_i + \lambda_i + \epsilon_t$$

where Y_t denotes GDP per capita in country at time t, $R_{d,t}$ it represents defense R&D, and $R_{c,t}$ captures civilian R&D. The variables K_t and L_t denote capital and labor inputs, respectively. Country fixed effects (α_1) control for time-invariant characteristics such as geography, institutional quality, and long-run industrial structure, while year fixed effects (λ_t) capture global economic conditions and common shocks affecting all countries.

To test whether the impact of defense R&D depends on domestic production capability, the model is extended to include an interaction term between defense R&D and exporter status:

$$lnY_t = \beta_0 + \beta_1 lnR_{d,t} + \beta_2 (lnR_{c,t} \times Exporter_i) + \beta_3 lnK_t + \beta_4 lnL_t + \alpha_i + \lambda_i + \epsilon_t$$

The variable $Exporter_i$ is a binary indicator equal to one for countries that are net exporters of defense technologies and zero for net importers. This specification allows the effect of defense R&D to vary across country types. The coefficient β_1 captures the effect of defense R&D in importer countries, while β_2 measures the additional effect for exporter countries. The total effect for exporters is given by $\beta_1+\beta_2$. Consistent with the theoretical framework, the expected relationship is that the effect of defense R&D will be stronger in countries with domestic production capability, where technological spillovers are more likely to be internalized.

In order to further explore the mechanisms identified in the theoretical framework, additional specifications incorporate variables designed to capture alternative channels. First, a war

dummy variable is introduced to account for periods of heightened geopolitical conflict. This variable takes the value of one during major conflict periods and zero otherwise. The inclusion of this variable allows the model to separate the effects of defense R&D from broader macroeconomic fluctuations associated with wartime conditions, which may independently affect economic performance.

Second, a financial variable, proxied by stock market capitalization, is included to test the capital accumulation channel. As discussed in the theoretical framework, defense R&D may increase expected profitability, which is reflected in higher firm valuations. Higher valuations can stimulate investment and contribute to capital accumulation, providing an indirect pathway through which defense R&D influences economic output. The inclusion of this variable allows the empirical analysis to examine whether financial markets play a role in transmitting the effects of defense-related innovation.

Finally, robustness checks are conducted by modifying the sample to account for structural differences across countries. In particular, resource-rich economies, especially major oil exporters, are excluded in some specifications. These countries may exhibit high levels of income driven primarily by natural resource revenues rather than innovation or production capacity, potentially biasing the estimated relationship between defense R&D and economic performance.

All regressions are estimated using fixed-effects models with standard errors clustered at the country level to account for within-country correlation over time. The use of fixed effects ensures that the estimated coefficients reflect within-country variation over time, rather than cross-country differences that may be driven by unobserved structural factors.

5. Empirical Results

This section presents the results of the empirical analysis examining the relationship between defense R&D and economic performance. The results are reported across multiple model specifications to evaluate robustness and to test the hypotheses developed in Section 4.2. Tables 2 through 4 summarize the regression results, while Figure 5 provides a visual representation of the estimated coefficients across specifications.

Figure 5: Regression Results

	(1) Base	(2) Int	(3) War	(4) V	(5) No SAU/Q
ln_rd	0.004	0.003	0.005	-0.003	0.003
	(0.005)	(0.006)	(0.006)	(0.007)	(0.006)
ln_rc	0.113**	0.113**	0.236***	0.125**	0.113**
	(0.043)	(0.043)	(0.037)	(0.051)	(0.043)
ln_K	0.428***	0.425***	0.478***	0.434***	0.425***
	(0.027)	(0.026)	(0.042)	(0.024)	(0.026)
ln_L	-0.732***	-0.716***	-0.440**	-0.617***	-0.716***
	(0.168)	(0.176)	(0.181)	(0.170)	(0.176)
ln_rd_x_exp		0.011	-0.017	0.012	0.011
		(0.018)	(0.016)	(0.017)	(0.018)
war			0.007		
			(0.007)		
ln_V				0.004	
				(0.020)	
N	660.000	660.000	660.000	490.000	660.000
R2 within	0.828	0.829	0.741	0.845	0.829

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The baseline specification, shown in Column (1) of Table 2, includes defense R&D along with controls for civilian R&D, capital, and labor, as well as country and year fixed effects. The model explains a substantial portion of the variation in GDP per capita, with an R^2 of 0.828. The

coefficient on defense R&D is 0.004 with a standard error of 0.005, and is statistically insignificant. This suggests that a one percent increase in defense R&D is associated with only a 0.004 percent increase in GDP per capita, an effect that is economically small and not statistically distinguishable from zero. In contrast, civilian R&D is positive and statistically significant (0.113), indicating a strong relationship between broader innovation activity and economic performance. Capital is also positive and highly significant (0.428), while labor enters with a negative and significant coefficient (-0.732).

Column (2) introduces the interaction between defense R&D and exporter status. The coefficient on defense R&D remains small and statistically insignificant (0.003), while the interaction term is positive (0.011) but also statistically insignificant. This suggests that, although the direction is consistent with the hypothesis that exporter countries may benefit more from defense-related innovation, the data do not provide statistically meaningful evidence of this effect. The explanatory power of the model remains largely unchanged (R^2=0.829).

Column (3) incorporates the war dummy variable to capture periods of heightened geopolitical conflict. The coefficient on defense R&D increases slightly to 0.005 but remains statistically insignificant. The interaction term becomes negative (-0.017) and remains insignificant, and the war variable itself (0.007) is also not statistically significant. The model fit declines (R^2=0.741), suggesting that the inclusion of wartime conditions does not improve explanatory power or alter the core results.

Column (4) introduces a financial variable, proxied by stock market capitalization, to test the capital accumulation channel. The model fit improves slightly (R^2=0.845), but the coefficient on defense R&D becomes slightly negative (-0.003) and remains statistically insignificant. The interaction term (0.012) and the financial variable itself (0.004) are also insignificant. These

results suggest that the financial channel does not translate into measurable differences in GDP per capita at the aggregate level.

Column (5) excludes major oil-producing countries, such as Saudi Arabia and Qatar, to address the concern that high GDP levels driven by natural resource wealth may obscure the underlying relationship. The results remain unchanged: defense R&D continues to be statistically insignificant (0.003), and the interaction term remains positive but insignificant (0.011). The R^2 returns to 0.829, consistent with the baseline model. This indicates that the lack of significance is not driven by resource-rich outliers.

Figure 6: Split-Sample regressions (Producers vs Importers)

Table 3. Split-Sample Regressions

	(1) Producers	(2) Importers (all)	(3) Importers (no oil)
ln(Defense R&D)	0.008	0.000	0.000
	(0.007)	(0.006)	(0.006)
ln(Civil R&D)	-0.034	0.115**	0.115**
	(0.035)	(0.055)	(0.055)
ln(Capital)	0.300***	0.430***	0.430***
	(0.055)	(0.026)	(0.026)
ln(Labor)	-0.089	-0.718***	-0.718***
	(0.241)	(0.177)	(0.177)
Observations	187	473	473
Countries	9	26	26
R2 within	0.917	0.844	0.844

Standard errors in parentheses
Country and year FE. SE clustered at country. * p<0.10 ** p<0.05 *** p<0.01
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6 presents robustness checks by replacing civilian R&D with total R&D (GERD). The results remain consistent across specifications. Civilian and total R&D continue to exhibit

positive and statistically significant relationships with GDP per capita, while defense R&D remains statistically insignificant. This confirms that the findings are not sensitive to how R&D is measured.

Figure 7: Coefficient Estimates Across Specifications.

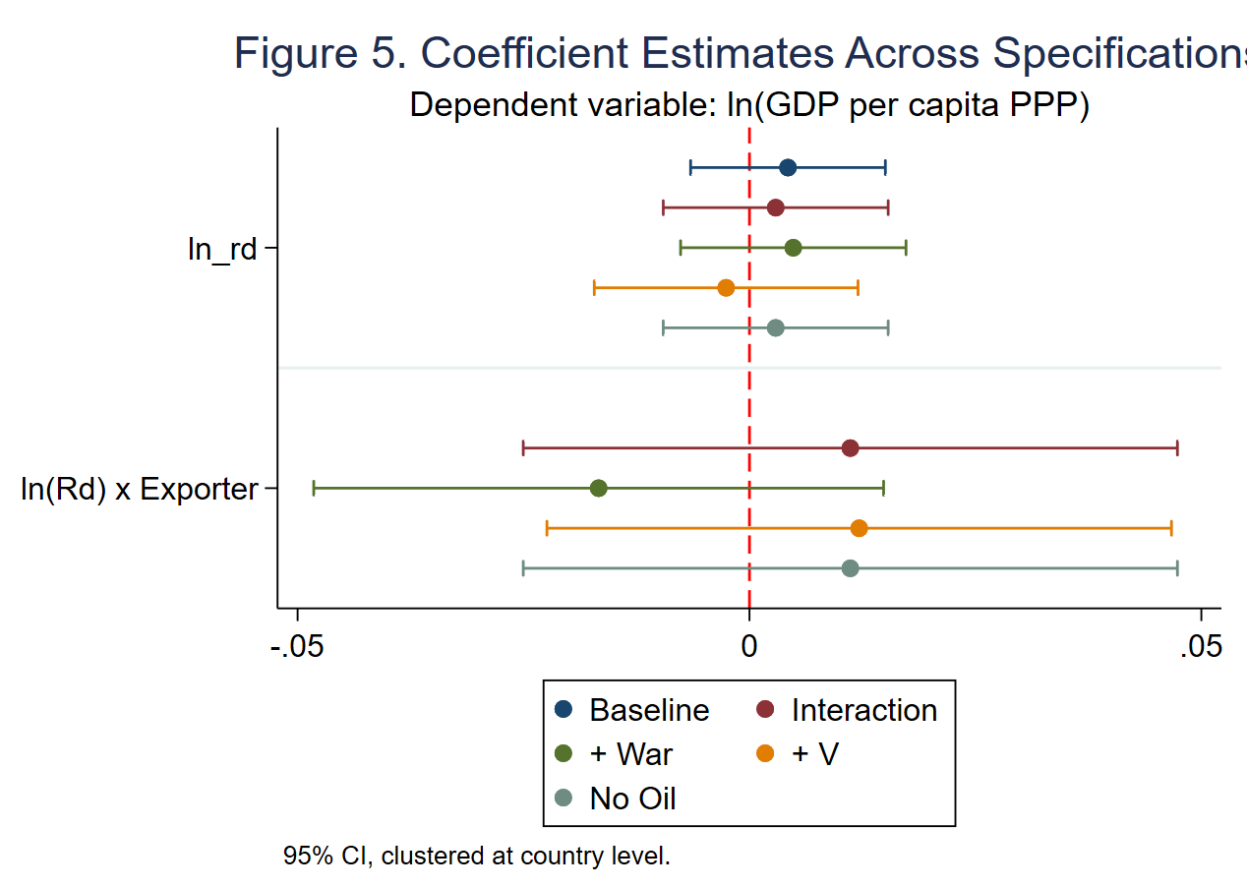


Figure 7 summarizes the estimated coefficients for defense R&D and its interaction with exporter status across all model specifications. The point estimates for defense R&D are consistently close to zero, and the confidence intervals overlap zero in all cases. Similarly, the interaction term displays wide confidence intervals that include zero across specifications. This visual evidence reinforces the regression results by showing that the estimated effects are not statistically distinguishable from zero.

Taken together, the empirical results do not support the hypothesis that defense R&D has a statistically significant effect on economic growth. While the theoretical framework suggests potential positive channels through productivity and capital accumulation, these effects are not observed in the aggregate data. Instead, the results indicate that the impact of defense R&D is either very small, highly context-dependent, or not captured by the available measures.

6.Conclusion

This paper set out to examine whether defense research and development (R&D) contributes to economic growth, and whether its effects differ across countries depending on domestic production capability. The theoretical framework suggested two primary channels through which defense R&D could influence economic performance: a productivity channel through technological spillovers and a capital accumulation channel through increased profitability and investment.

However, the empirical results do not provide statistically significant evidence supporting this relationship. Across all model specifications, including interaction terms, war controls, financial variables, and sample restrictions, the estimated effect of defense R&D on GDP per capita remains small and statistically insignificant. While some coefficients are positive and consistent with the theoretical framework, they are not precisely estimated and cannot be distinguished from zero. In contrast, civilian and total R&D consistently exhibit positive and significant relationships with economic performance, suggesting that broader innovation activity plays a more robust role in economic growth.

These findings highlight important limitations in the available data. Defense R&D is inherently difficult to measure due to its classified nature and the lack of consistent, transparent reporting

across countries. Unlike civilian R&D, which is relatively well-documented, defense-related innovation is often embedded within government budgets, military programs, and private contracts that are not fully observable. As a result, the variable used in this analysis may not fully capture the true scale or impact of defense-related technological activity.

In addition, the dataset is limited to the period from 2000 to 2022. While this captures important geopolitical events, it may not fully reflect more recent developments in global defense strategy and technological competition. For example, emerging tensions and strategic competition, including recent developments involving Iran and broader shifts in U.S. defense policy, may play a significant role in shaping future defense-related innovation and its economic effects. Extending the dataset to include more recent observations could provide additional insight into these dynamics.

Despite the lack of statistically significant results, this does not necessarily imply that defense R&D has no economic impact. Existing research suggests that defense-related R&D can generate spillover effects and stimulate private innovation, particularly in industries such as aerospace and advanced manufacturing . However, these effects may be highly context-dependent, industry-specific, or realized over longer time horizons that are not captured in aggregate cross-country data.

From a broader perspective, the findings of this paper suggest that the relationship between defense R&D and economic growth is more complex than a simple positive or negative effect. It may depend critically on how defense spending is structured, whether it is integrated into domestic industries, and how effectively technological spillovers are absorbed into the broader economy.

For future research, a more detailed analysis using firm-level or industry-level data may provide a clearer understanding of the mechanisms at play. In particular, examining how defense

contracts influence private-sector innovation, investment behavior, and productivity could help uncover relationships that are not visible in aggregate macroeconomic data.

Personally, I believe that defense R&D plays a meaningful role in shaping long-term economic and technological development. While this paper does not find strong statistical evidence at the aggregate level, the theoretical channels and existing literature suggest that such a relationship likely exists under the right conditions. In a world characterized by geopolitical uncertainty, the idea of *“si vis pacem, para bellum”*—if you want peace, prepare for war—remains relevant. This does not imply support for conflict, but rather emphasizes the importance of preparedness and strategic investment. Ideally, we would live in a world without conflict, but given current global conditions, investment in defense R&D may serve not only as a tool for national security but also as a driver of technological progress and economic resilience.

References

lock, Fred L., and Matthew R. Keller, eds. *State of Innovation: The U.S. Government's Role in Technology Development*. Boulder, CO: Paradigm Publishers, 2011.

Deleidi, M., C. Fontanari, and S. J. Gahn. "Autonomous Demand and Technical Change: Exploring the Kaldor–Verdoorn Law on a Global Level." *Economics and Politics* 40 (2023): 57–80. https://doi.org/10.1007/s40888-023-00294-y.

Draca, Mirko. "Reagan's Innovation Dividend? Technological Impacts of the 1980s US Defense Build-up." Working Paper. Coventry, UK: University of Warwick, 2013.

Dunne, Paul J., and Mehmet Uye. "Defence Spending and Development." *ResearchGate*, January 2009. https://www.researchgate.net/publication/241768513_Military_Spending_and_Development.

Harutyunyan, Gayane E. "The Impact of Military R&D on the Innovative Development of the Civilian Sector." *Public Administration Issues*, Special Issue (2017): 27–37.

Inal, V., T. Gurdal, T. Degirmenci, and M. Aydin. "The Effects of Military Expenditures on Labor Productivity, Innovation and Economic Growth for the Most Militarized Countries: Panel Data Analysis." *Kybernetes* 53, no. 3 (2024): 821–840. https://doi.org/10.1108/K-06-2022-0852.

Mazzucato, Mariana. *The Entrepreneurial State: Debunking Public vs. Private Sector Myths*. London: Anthem Press, 2013.

McDonough, Matthew David. "The Effects of Defense R&D on Technological Progress: Resource Diversion versus Spillover Benefits." University of Texas at Dallas, 2012.

Moroz, David. “On What Topics Does the Scientific Community Cooperate in Defense Economics? A Science Mapping of the First 30 Years of the Journal Defence and Peace Economics.” *Defence and Peace Economics* 36, no. 4 (2025): 555–576.

Peretto, Pietro F. “Government Spending and Growth.” Working Paper. Durham, NC: Duke University. https://public.econ.duke.edu/~peretto/CPW%20Government%20Spending%20and%20Growth.pdf.

Segarra, Elena, and Jorge Uxó. “Demand-Led Growth, the Supermultiplier, and Fiscal Policy: A Review of the Literature and Some Applications to the European and Spanish Context.” *National Accounting Review* 7, no. 4 (2025): 476–500. https://doi.org/10.3934/NAR.2025020.

University of Sydney. “ECON 2011-2.” School of Economics. https://ses.library.usyd.edu.au/bitstream/handle/2123/7214/ECON%202011-2.pdf.